# Data Request Tables

## Data Request and Prediction-Point Requirement

The proposed Data Science project will predict whether an order will be returned using information available at or before the point of purchase. The requested data should therefore include sufficient historical information to construct predictive features while also providing the return outcome required to train and evaluate the classification model.

All predictive variables must represent information available at the prediction point. Variables generated after purchase, delivery or return initiation must not be used as model predictors. Such variables may still be requested where they are required for target construction, model evaluation or post-hoc analysis. Historical behavioural, seller and operational measures should be calculated using information available before the prediction point to prevent data leakage.

## Table 1: Orders and Transactions

**Important:** Order-level variables must reflect information available at the time the order was placed.

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| Order ID | String | ORD849302 | Unique identifier for each order |
| Customer ID | String | CUST29482 | Pseudonymised customer identifier that can be linked across datasets |
| Order date/time | Datetime | 2026-04-15 14:32:10 | Use consistent yyyy-MM-dd HH:mm:ss format |
| SKU | String | SKU183920 | Unique product identifier. If an order contains multiple SKUs, provide order-item records linked using Order ID. |
| Seller ID | String | SELL4839 | Marketplace seller identifier; identify first-party orders appropriately |
| Quantity | Integer | 2 | Number of units purchased |
| Unit price | Numeric | 899.00 | Price in South African rand |
| Promotion amount | Numeric | 100.00 | Discount applied to order |
| Payment method | Categorical | Card | Use a controlled list of payment methods |
| Delivery option | Categorical | Standard | Use a controlled list |
| Customer province | Categorical | GP | Use South African province codes |

## Table 2: Returns

The Returns dataset is required primarily to construct the target variable, evaluate model performance and analyse the characteristics and costs of returned orders. Post-purchase return information must not be used as input features for the predictive model.

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| Return ID | String | RET55081 | Unique identifier for each return event. |
| Order ID | String | ORD849302 | Links the return to the original order. |
| Return date/time | Datetime | 2026-04-20 09:15:00 | Date/time the return was initiated. If available, also provide return receipt date/time separately. Post-purchase; not a predictive feature. |
| Returned quantity | Integer | 1 | Number of units returned. Used for outcome analysis/target construction; not a predictive feature. |
| Return reason | Categorical | Wrong size | Standardised return reason. Known only after return and must not be used as a predictive feature. |
| Return condition | Categorical | Unopened | Controlled list of return conditions. Post-return information; not a predictive feature. |
| Refund amount | Numeric | 899.00 | Refund amount in ZAR. Post-purchase information; not a predictive feature. |
| Refund outcome | Categorical | Approved | Controlled refund outcome. Post-purchase information; not a predictive feature. |
| Return cost | Numeric | 65.00 | Processing/reverse-logistics cost in ZAR. Useful for business impact analysis; not a predictive feature. |
| Avoidable return flag | Binary | 1 | Existing business-defined classification if available. If not available, definition should be agreed with STADIOalot. Not a predictive feature unless known at prediction time. |

## Table 3: Product Catalogue

**Important:** Product ratings and review counts must reflect information available before the order was placed, not information added afterwards.

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| SKU (Product ID) | String | SKU183920 | Unique product identifier used to link product catalogue information to orders and inventory records. |
| Product category | Categorical | Apparel | Use standard category hierarchy |
| Subcategory | Categorical | Women's shoes | Standardised subcategory |
| Brand | Categorical | Brand A | Consistent brand identifier |
| Size | Categorical | Large | Required where applicable |
| Colour | Categorical | Black | Standardised colour |
| Product price | Numeric | 899.00 | Historical product price at the time of purchase. |
| Product rating | Numeric | 4.3 | Rating available at the prediction point. Document the rating scale, e.g. 0–5. |
| Review count | Integer | 128 | Number of reviews available before the prediction point. |
| Listing age days | Integer | 84 | Number of days the product had been listed at the prediction point. Must be ≥ 0. |

## Table 4: Marketplace Sellers

**Important:** Seller-performance variables must be calculated using historical information only. The current order must not contribute to these measures.

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| Seller ID | String | SELL4839 | Unique seller identifier |
| Seller rating | Numeric | 4.2 | Seller rating available at the prediction point. Document the rating scale. |
| Seller tenure months | Integer | 18 | Time seller has been active |
| Average shipping days | Numeric | 2.8 | Historical average shipping time calculated using orders before the prediction point. |
| Late shipment rate | Numeric | 0.07 | Historical proportion of previous shipments classified as late, calculated before the prediction point. |
| Complaint count | Integer | 12 | Number of seller complaints during a defined historical period before the prediction point. |

## Table 5: Customer Information

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| Customer ID | String | CUST29482 | Pseudonymised unique customer identifier; must link to the Orders and Customer Behaviour datasets |
| Customer age | Integer | 25 | Age of customer at the time of the order |
| Customer gender | Categorical | Female | Use controlled values; include an Unknown/Not disclosed category where applicable |
| Customer province | Categorical | GP | South African province code associated with the customer's location at the prediction point. |
| Account creation date | Datetime | 2023-06-14 10:25:00 | Date and time the customer account was created |
| Customer tenure months | Integer | 36 | Customer tenure calculated as at the prediction point. |
| Preferred shopping channel | Categorical | Mobile App | Controlled values such as Website, Mobile App, or Other |
| Average order value | Numeric | 1,245.50 | Historical average order value calculated using eligible orders before the prediction point, in ZAR. |
| Total historical spend | Numeric | 38,615.50 | Cumulative customer spend before the prediction point, in ZAR. |
| Previous returned orders | Integer | 2 | Number of orders returned by the customer before the prediction point. |

## Table 6: Customer Behaviour

**Important:** Behavioural variables must be measured using a clearly defined look-back period and must exclude activity associated with the current order after the prediction point.

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| Customer ID | String | CUST29482 | Pseudonymised identifier linking behaviour to the customer. |
| Look-back period | Integer | 30 | Number of days over which behavioural activity is measured. The period should end at or before the prediction point. |
| Sessions | Integer | 24 | Number of shopping sessions during the defined look-back period before purchase. |

| | | | |
|---|---|---|---|
| Product views | Integer | 183 | Number of product detail views during the look-back period before purchase. |
| Add-to-basket count | Integer | 19 | Number of add-to-basket events during the look-back period before purchase. |
| Previous orders | Integer | 31 | Number of completed orders before the prediction point. |
| Previous returns | Integer | 6 | Number of previous returned orders before the prediction point. |
| Customer return rate | Numeric | 0.19 | Previous returns divided by eligible previous orders. Calculated using information available before the prediction point. |

## Table 7: Inventory and Warehouse

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| SKU | String | SKU183920 | Product identifier linked to the order/product catalogue. |
| Distribution centre | Categorical | DC3 | Distribution centre associated with fulfilment. Where possible, provide the centre known/assigned at the prediction point. |
| Stock on hand | Integer | 240 | Units available at the prediction point. Must be ≥ 0. |
| Stock age days | Integer | 46 | Age of available stock at the prediction point. Must be ≥ 0. |
| Historical pick/pack time | Numeric | 7.5 | Historical average pick/pack time, preferably calculated from previous orders before the prediction point. Unit: minutes. |
| Inbound date | Datetime | 2026-03-01 08:00:00 | Date/time the relevant stock was received into the distribution centre. Must be available at or before the prediction point. |

## Table 8: Delivery and Logistics

| Column name | Data type | Example | Additional information |
|---|---|---|---|
| Order ID | String | ORD849302 | Links logistics information to the order. |
| Estimated delivery date/time | Datetime | 2026-04-18 16:20:00 | Estimated delivery information available at the time of purchase. Can be used as a predictive feature if available. |

| | | | |
|---|---|---|---|
| Delivery option | Categorical | Standard | Delivery option selected at purchase. |
| Delivery province | Categorical | KZN | Destination province associated with the order. |
| Route distance km | Numeric | 18.4 | Estimated/recorded route distance available at the prediction point, where applicable. |
| Historical delivery time | Numeric | 2.4 | Historical average delivery duration for the relevant route/region, calculated before the prediction point. Unit: days. |
| Historical delivery failure rate | Numeric | 0.06 | Historical proportion of failed/late deliveries calculated before the prediction point. |
| Actual delivery date/time | Datetime | 2026-04-19 11:45:00 | Actual delivery timestamp. Post-purchase; use for evaluation/analysis only, not prediction. |
| Delivery attempts | Integer | 2 | Number of attempts for the current order. Post-purchase; not a predictive feature. |
| On-time flag | Binary | 0 | 1 = on time, 0 = late. Post-purchase; not a predictive feature. |

**Target Variable**

The target variable for the eventual Data Science project will be:

| Target variable | Value | Meaning |
|---|---|---|
| Returned | 1 | The order meets the agreed definition of a returned order within the specified observation window. |
| Returned | 0 | The order does not meet the agreed definition of a returned order within the specified observation window. |

This creates a binary classification problem.

The unit of prediction is the order. The return observation window and definition of an order being “returned” should be agreed with STADIOalot before model development.

For orders containing multiple items, the project should establish whether an order is classified as Returned = 1 when at least one item is returned, or according to another business-approved aggregation rule.